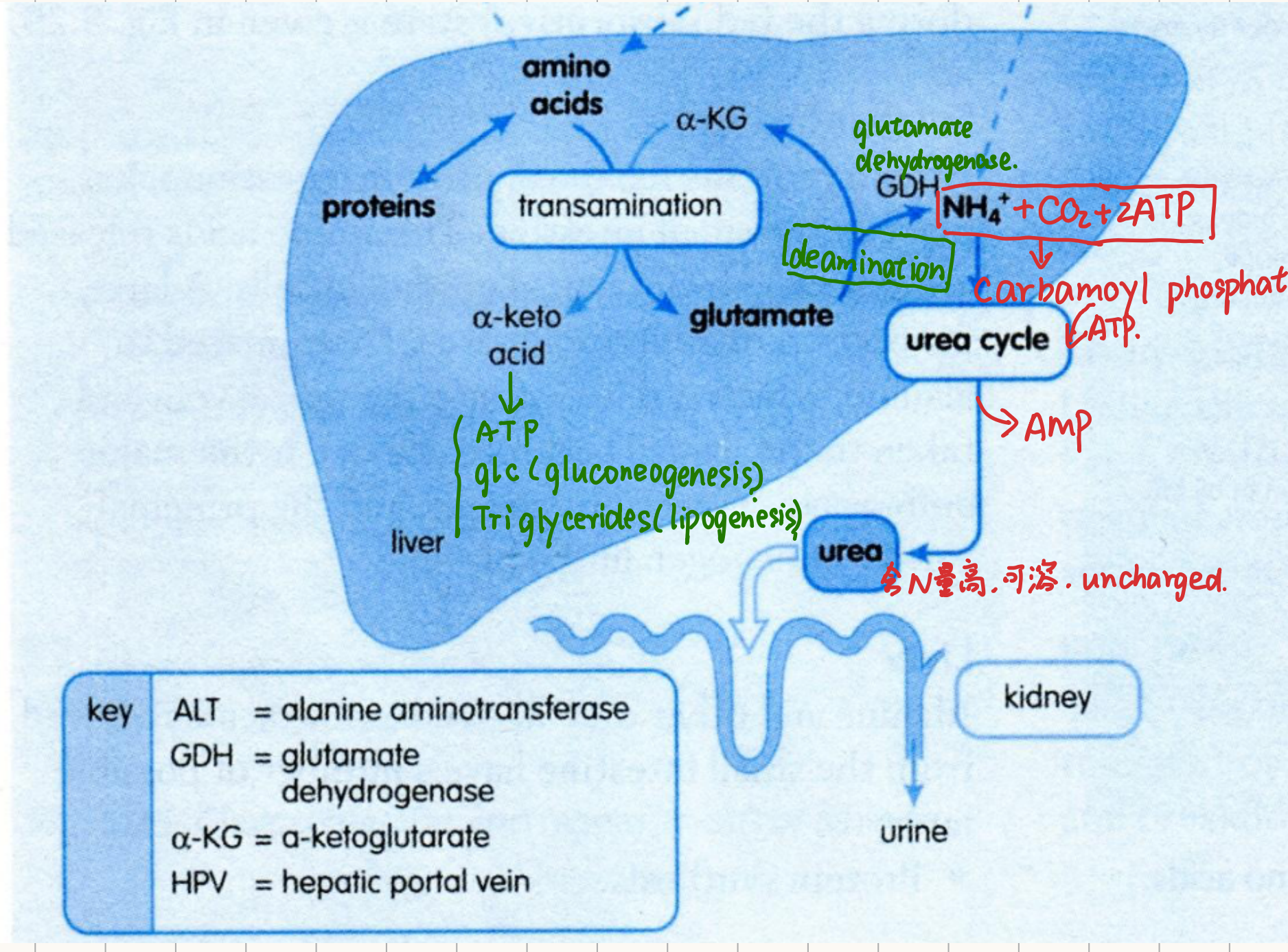
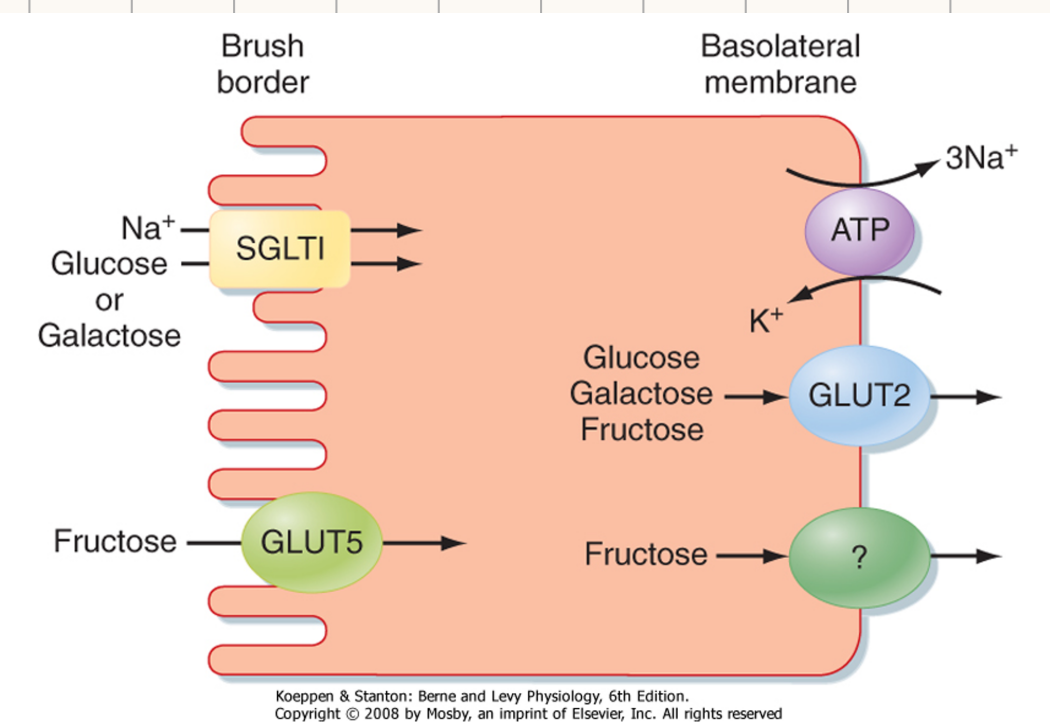


消化吸收过程:		
口	protein	lingual lipase: 短链脂肪, 在胃中活跃 (dorsal surface of tongue secretion)
胃	pepsin $\xleftarrow[\text{autocatalytic}]{\text{HCl (parietal)}}$ pepsinogen (chief cell) large polypeptide chain.	gastric lipase: 中长链脂肪 (胃里在肠激活) (chief cell)
胰	enteropeptidase (intestinal enterocyte) $\rightarrow$ trypsinogen $\xrightarrow{\text{auto}}$ trypsin chymotrypsin, carboxypeptidase $\xleftarrow{\text{trypsin}}$ chymotrypsinogen, pro carboxypeptidase small peptides.	fat globule. bile salt: detergent small fat droplet. pancreatic lipase: 脂肪酶. 水解为 FA + monoglyceride. colipase: bind to lipase. 结合到脂肪上. cholesterol esterase: 水解胆固醇酯键. phospholipase A₂: 水解磷脂.
小肠	aminopeptidase (enterocyte brush border) intracellular peptidase.	无 无
吸收	aa Na⁺-aa transporter aa transporter blood. protein 吸收: duodenum, jejunum	Micelles with monoglyceride + FA. diffuse 重新合成 triglyceride. protein chylomicrons 进入淋巴管
相关疾病	① pancreatic insufficiency ② Hartup: 无法转运中性 aa ③ Cystinuria: 无法转运含巯基酸性 aa	① pancreatic insufficiency: steatorrhea. Pro 和 CHO 较 Lipid 受累程度更小. 可通过补充短链 FA 治疗.
其余相关代谢过程		primary CHO malabsorption: 遗传 secondary - - - -: 因其余疾病导致小肠上受累 lactose intolerance: lactose 在大肠中被细菌分解为 H₂, CO₂, SCFA. gas $\rightarrow$ distension, cramp, flatulence SCFA $\rightarrow$ acidic pH. water retention $\rightarrow$ osmotic diarrhea.



lipogenesis (liver, enterocyte, adipose): FA + glycerol $\rightarrow$ 脂肪
 lipolysis: adipose: 受激素刺激而分解: FFA + glycerol.
 肝: 肝
 如: 如
 糖: 糖
 解: 解
 β -oxidation

脂质运输:

① exogenous pathway (处理从外界摄入的脂质):
 小肠合成的TGA, 摄入的磷脂, 胆固醇 + ApoB48
 $\downarrow$ ApoC, ApoE (来自HDL)
 CM
 $\downarrow$ ApoC激活血管内皮lipoprotein lipase (LPL), 释放FA供其他组织利用
 结合肝 remnant receptor, 被肝摄入.

② endogenous pathway (将肝合成的脂肪供其他组织):
 VLDL (肝合成) $\xleftrightarrow[\text{HDL给VLDL ApoC}]{\text{VLDL给HDL 胆固醇}}$ HDL (肝, 小肠合成).
 $\downarrow$ LPL $\rightarrow$ FFA
 IDL $\rightarrow$ 肝降解.
 LDL (ApoC, ApoB100, 胆固醇)
 $\frac{1}{3}$: 被 phagocyte 吞噬.
 $\frac{2}{3}$: 进入各 cell 中, 释放胆固醇.

VLDL, IDL, LDL: bad cholesterol. $\Rightarrow$ cause atherosclerosis
 HDL: good cholesterol.

Apolipoprotein	
Structural components	Solubilize the lipids
Recognition sites or ligands for receptors	
Activators or coenzymes for enzymes involved in lipid metabolism	

Functions of the major apolipoproteins	
Apolipoprotein	Function
A-I	activates lecithin:cholesterol acyl transferase
A-II	activates hepatic lipase
B-48	structural: chylomicrons (CM)
B-100	structural: binds to the apoB/E (LDL) receptor increases cholesterol uptake
C-I	cofactor for lecithin:cholesterol acyl transferase
C-II	activates lipoprotein lipase
C-III	inhibits lipoprotein lipase?
E	binds to apoB/E (LDL) receptor and increases uptake of LDL and remnant particles including CM remnants

Fig. 4.22 Functions of major apolipoproteins.

Lipoprotein: function				
Classification and properties of lipoproteins				
Class	Main composition	Diameter (nm)	Source and function	Major apolipo-proteins
CM	90% triacylglycerol	500	transport of dietary triacylglycerol	A-I, II, B-48, C-I, II, III, E
VLDL	65% triacylglycerol	43	transport of endogenously synthesized triacylglycerol from the liver to peripheral tissues	B-100, C-I, II, III, E
IDL	35% phospholipid 25% cholesterol	27	formed by partial breakdown of VLDL, precursor of LDL	B-100, C-III, E
LDL	50% cholesterol 25% protein	22	formed by breakdown of IDL; carries cholesterol to peripheral tissues	B-100
HDL	55% protein 25% phospholipid	8	formed in the liver and intestine; 2 main functions: - reverse cholesterol transport: removes cholesterol from tissues and takes it to liver, 'cholesterol scavenger' - exchanges apolipoproteins and cholesterol esters with chylomicrons and VLDL	A-I, II, C-I, II, III, D, E